



25 W continuous-wave Yb:LiLuF₄ single-crystal-fiber laser oscillator

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ABSTRACT

We present the first demonstration of a solid-state laser oscillator based on a Yb-doped, birefringent single crystal fiber grown by the micro-pulling-down method. Pumping a 42-mm-long, 2%-at. Yb³⁺:LiLuF₄ crystal with a 105-W maximum power fiber coupled laser diode, we obtained a maximum output power of 10.4 W, with diffraction limited beam quality ($M^2 < 1.1$), in a simple plane-concave resonator. By modifying the resonator layout to increase the laser efficiency, we obtained output power up to 25 W, with still fairly good beam quality ($M^2 \approx 2$). Thermal lensing effect and laser intrinsic slope efficiency and losses were characterized in both resonator configurations.

1. Introduction

Single crystal fibers (SCFs) grown by the micro-pulling-down (μ -PD) have proven to be a practical solution for power scaling of solid-state lasers, their main features being a long absorption length and a favorable surface-to-volume ratio that enhances thermal dissipation. μ -PD permits to obtain SCF with diameters from few to ~ 1 mm. Thinner SCFs (down to 0.1 mm of diameter) can be grown with the more complex Laser Heated Pedestal Growth (LHPG) method [1]. Such small diameter is strictly required only to exploit guiding effect for the pump and the laser beam. However, the availability of bright, high-power laser diodes permits to obtain excellent results also with larger SCF grown by the μ -PD. Incidentally, although LHPG can produce the small SCFs diameters, it is not suited to fluoride growth.

To now, crystal growth with μ -PD technique was successfully employed mainly for cubic crystals and in particular YAG. Oscillators operating in continuous-wave (cw) regime based on such crystalline host doped with Nd³⁺ [2,3] and Yb³⁺ [3,4] for laser emission around 1 μ m, and with Ho³⁺ at 2 μ m [5], have been reported in the last decade. Owing to intrinsic advantages such as low quantum defect, broad absorption bandwidth and the absence of excited state absorption, the most impressive results were obtained with Yb:YAG both for oscillators [4] and for amplification of pulses in the picosecond [6] and femtosecond regime [7].

We recently demonstrated for the first time, successful growth of a birefringent Yb-doped LiLuF₄ (LLF) SCF and cw-regime high-power amplification with excellent beam quality preservation even at very high

(~ 90 W) absorbed pump power [8]. Birefringent SCFs are particularly attractive, as they provide a quite straightforward means of avoiding beam quality degradation at high thermal load due to depolarization. LLF crystalline host is isomorph to LiYF₄ (YLF), but grows congruently, avoiding polycrystals formation during the fabrication process. When doped with Yb³⁺, LLF has absorption/emission cross sections and fluorescence lifetime practically indistinguishable from YLF [9], but thanks to a better matching of Lu³⁺ and Yb³⁺ ion atomic mass and radius, thermal conductivity in LLF is expected to be almost independent on Yb³⁺ concentration as for other hosts that can be grown alternatively with Yttrium or Lutetium [10,11]. The sample under investigation was grown along the *a*-axis. Details on the crystal growth and orientation of the sample employed in the experiments are reported in [8].

In this work we present what is, to the best of our knowledge, the first demonstration of efficient laser oscillation based on a Yb-doped birefringent SCF. The 42-mm-long, ~ 2.8 -mm diameter, 2%-at. Yb-doped LLF SCF we employed in our aforementioned amplification experiments was here tested as active medium of compact cw resonators. Employing a simple plane-concave resonator layout, we demonstrated up to 10 W, with almost diffraction limited beam quality at about 95 W absorbed pump power. By enhancing the efficiency of the laser through a better matching between the pump and the resonator mode, we obtained up to 25 W output power with slope efficiency approaching 40% with the optimum $R = 50\%$ output coupler (OC) and still good beam quality ($M^2 \sim 2$). Intrinsic laser efficiency, resonator losses (comprising

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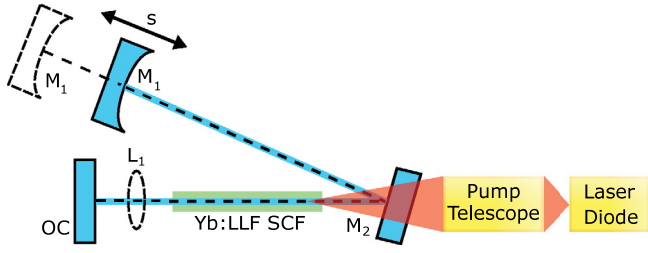


Fig. 1. Laser layouts. M_1 : Spherical mirror $R = -200$ mm, HR at 1020 nm; M_2 : plane dichroic mirror HR at 1020 nm, HT at 976 nm; L_1 : spherical lens $f = 40$ mm, AR coated at 1020 nm; OC: Plane output coupler mirror; s : displacement of the mirror M_1 .

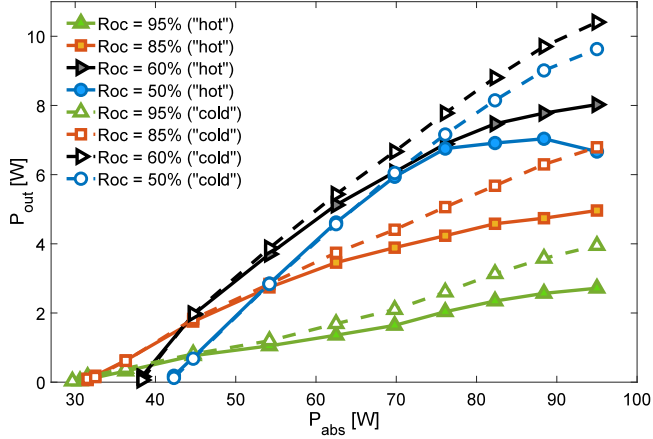


Fig. 2. Output power versus absorbed pump power for different R_{OC} in the “hot” (solid lines) and “cold” condition (dashed lines) for the plane-concave resonator.

SCF losses) and thermal lensing effect are investigated in both cavity configurations. Our results very well compare with those reported for cubic crystals μ -PD grown SCF at a similar developmental stage of the technology [3], showing the excellent potential of birefringent fluoride SCFs.

2. Experiments

The setup of the experiments is shown in Fig. 1. The Yb:LLF SCF was wrapped in indium foils and soldered in a copper mounting that was water cooled at 10°C during the experiments. As in the amplification studies reported in [8], the pumping unit was a multi-mode fiber-coupled laser diode ($NA = 0.22$, $\varnothing = 105 \mu\text{m}$), providing a maximum power of 105 W at 976 nm. The maximum absorbed power under lasing conditions was 95 W. The pump beam was coupled in the gain medium through an achromatic lenses telescope yielding a pump spot radius $w_p \approx 200 \mu\text{m}$ in the Yb:LLF SCF. The focusing lens of the pump telescope was mounted on a micrometric stage to precisely control the position of the pump beam waist inside the active medium. The laser crystal had uncoated, in-house optically-polished end-facets. Given the LLF refractive index $n_e \approx 1.47$ at 1020 nm, the Fresnel loss at the air/crystal interface is $\sim 3.6\%$.

At first, we set up a simple plane concave resonator (light blue, solid-line setup in Fig. 1), with a total physical length of 162 mm, the OC mirror being placed at about 10 mm from the end facet of the SCF. From the ABCD simulation of the fundamental cavity mode, we estimated a beam waist on the OC $w_{oc} \approx 170 \mu\text{m}$, resulting in a reasonably collimated cavity mode inside the active medium with a maximum $1/e^2$ radius $w \approx 185 \mu\text{m}$ at the pumped facet of the SCF.

In Fig. 2 we present the output power (P_{out}) as a function of the absorbed pump power (P_{abs}), for different values of the OC reflectivity

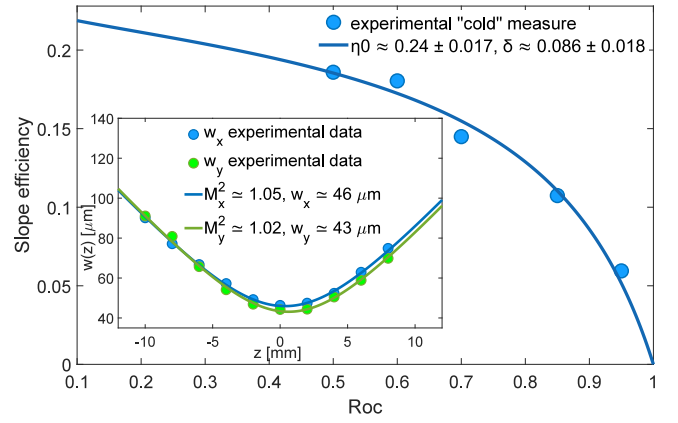


Fig. 3. Caird analysis performed on the plane-concave resonator in “cold” conditions. In the inset we show the beam quality measurement at the maximum laser output power.

(R_{oc}) ranging from 95% to 50%. Laser emission was polarized along the SCF c -axis (that was oriented along the y -axis of the lab reference, perpendicular to the optical table) and occurred at 1020 nm, close to the peak of the emission cross section [9], for all the OC reflectivities. As in the case of Ref. [8], measurements were performed both in “hot” conditions, i.e., waiting the thermal stabilization at each different P_{abs} (full dots solid lines in Fig. 2), and in “cold” conditions, i.e., by turning off the pump diode module to cool the SCF to about 10°C, and then by measuring the output power immediately after turning the pump diode on (empty dots, dashed lines in Fig. 2). In both conditions, the optimum coupler was $R_{oc} = 60\%$, yielding a maximum $P_{out} = 8$ W and $P_{out} = 10.4$ W in “hot” and “cold” regime, respectively. By comparing the performance of the laser in the two different regimes, we can observe a significant roll-off of the slope of the curves for $P_{abs} \geq 60$ W, in fairly good agreement with a similar degradation of the performance already observed in amplification experiments [8]. The absorbed pump power at laser threshold was $P_{abs,th} \approx 30$ W for the $R = 95\%$ OC and increased only to $P_{abs,th} \approx 42$ W with the $R = 50\%$. This is a clear indication that reabsorption losses at the 1020 nm laser wavelength are the dominant factor in fixing the laser threshold.

To assess the spatial beam quality, the laser beam was collimated with a $f_1 = 400$ mm spherical lens and focused with a $f_2 = 75$ mm spherical lens. A CCD camera was scanned along the focal plane of the telescope. In the inset of Fig. 3, we show the result of the measurement carried out with the optimum OC at the maximum thermal load ($P_{abs} \approx 95$ W). The beam was almost circular and diffraction limited with $M_x^2 \times M_y^2 \approx 1.05 \times 1.02$. From the magnification factor of the telescope we used for the measurements, it is possible to retrieve a $1/e^2$ radius on the OC mirror for the embedded TEM_{00} mode $w_{x,HP} \times w_{y,HP} \approx 239 \mu\text{m} \times 227 \mu\text{m}$. The same measurement was repeated at an almost halved thermal load (at $P_{abs} \approx 45$ W) yielding the same M^2 and a retrieved embedded TEM_{00} mode radius $w_{x,LP} \times w_{y,LP} \approx 208 \mu\text{m} \times 211 \mu\text{m}$. From these data, through ABCD modeling of the resonator, we could estimate the focal length of the thermal lens f_{th} in the SCF. In Fig. 4 we show the evolution of the beam waist radius on the OC for different values of f_{th} as a function of the displacement s with respect to the nominal distance between the curved mirror M_1 and the pumped facet of the active medium (see Fig. 1). As it can be seen, in this resonator configuration, w_{oc} is quite insensitive to the exact position of M_1 , which we could measure within an uncertainty of ± 0.5 mm. As expected, due to the different thermal conductivity and temperature coefficient of the refractive index of Yb:LLF [12], the thermal lens is moderately astigmatic with a focal length $f_{th,x} \times f_{th,y} \sim -115 \text{ mm} \times -125 \text{ mm}$, in fairly good agreement with the value estimated in similar pumping conditions and with similar laser beam radius in the SCF in our previous investigations [8].

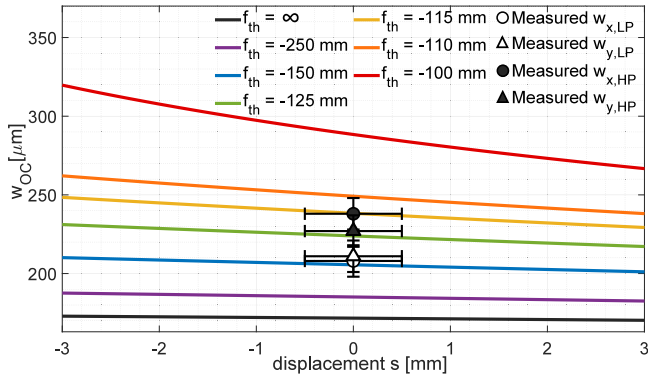


Fig. 4. Evolution of the plane-concave resonator fundamental mode beam waist radius on the OC as a function of the displacement s of mirror M_1 from the nominal distance of 110 mm from the pumped facet of the SCF and for different values of the focal length of the thermal lens in the SCF. Empty circle (triangle) corresponds to the measured w_{oc} along the x (y) direction of the lab reference at low P_{abs} (LP). Solid circle (triangle) corresponds to the measured w_{oc} along the x (y) direction of the lab reference at high P_{abs} (HP).

To determine cavity round trip losses δ and laser intrinsic slope efficiency η_0 (accounting for quantum efficiency and mode-matching efficiency between pump and cavity mode along the active medium), we performed a Caird analysis [13], by fitting the slope η of the output power characteristic curves in the “cold” condition as a function of R_{oc} . Given the large value of the OC transmission, we fit the data with the Rigrod equation [14]:

$$\eta = \eta_0 \frac{1 - R_{oc}}{1 - R_{oc} + \sqrt{R_{oc}} \left(\frac{1}{1 - \delta} - 1 + \delta \right)} \quad (1)$$

The result of the Caird analysis carried out on the performance of the plane-concave resonator is shown in Fig. 3. For what concerns the resonator losses $\delta \approx 8.6\%$, the largest contribution can be reasonably attributed to the residual losses at the uncoated crystal/air interfaces. Indeed, the imperfect parallelism of the SCF facets, permits only partial cancellation of the Fresnel reflection. The best fit intrinsic slope efficiency $\eta_0 \approx 24\%$ is rather low. E.g., if we compare this result with our previous preliminary investigations on shorter Yb:YLF crystals fabricated with the micro-pulling-down technique [15] in the same growing apparatus, we notice a significant reduction, of about a factor 2. The more likely reason of such a major drop is related to a poor mode matching between the cavity fundamental mode and the pump mode along the whole SCF length, due to the different divergence of the pump and the cavity mode. Experimentally, we observed that the optimization of the output power required the pump beam to be focused closer to the pumped crystal facet. Therefore, a good mode matching between the pump beam and the rather large fundamental resonator mode is realized only in a relatively short region of the SCF, with the end-part of the active medium likely enhancing fundamental mode selection due to reabsorption losses on the tails of the pumped region.

To overcome this major limitation, we decided to modify the resonator to reduce the beam waist in the active medium and try to better match the pump beam divergence in order to increase the overlap between cavity and pump mode along the whole SCF length, basically trading off fundamental mode selection for a better laser efficiency. The more straightforward way to obtain this effect was to introduce a spherical lens (L_1 in Fig. 1) between the end facet of the crystal and the output coupler (see Fig. 1, black dashed line), and to adjust the physical length of the resonator, in order to obtain an almost confocal cavity layout. The mirror M_1 (see Fig. 1) was mounted on a translation stage in order to be able to finely tune its distance from the pumped facet of the active medium, since it determines the position of the cavity waist

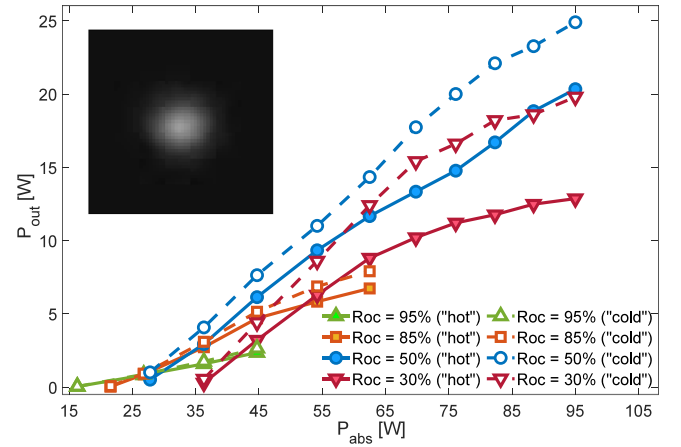


Fig. 5. Output power versus absorbed pump power for different R_{oc} in the “hot” (solid lines) and “cold” condition (dashed lines) for the confocal resonator. In the inset it is shown the smooth beam profile at the maximum output power.

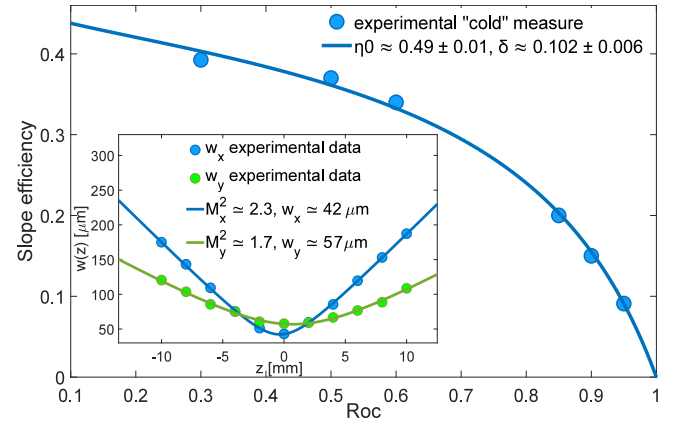


Fig. 6. Setup 2 Rigrod fitting curve in the “cold” conditions. The inset shows the beam quality measurement with $R = 50\%$ OC.

inside the SCF. The best results were obtained with a total resonator physical length of ~ 230 mm. In this condition, according to the ABCD modeling, the fundamental cavity mode waist inside the SCF has a $1/e^2$ radius $w_0 \approx 85 \mu\text{m}$.

The experimental results are shown in Fig. 5. Given the significant increase of the output power, in order to preserve the in-house-polished SCF facets, we limited the intracavity power to ~ 100 W, therefore, when we employed higher reflectivity OCs, we did not use all the available pump power.

The maximum output power was 20 W in “hot” conditions with the optimum $R_{oc} = 50\%$, and it increased up to 25 W in “cold” condition. The laser threshold was greatly lowered for all the OCs reflectivities, suggesting a significant reduction of the reabsorption losses. This can be explained considering that the fundamental cavity mode, which is the first to reach the threshold, thanks to its smaller dimension, better overlaps with the more inverted region of the SCF throughout all the length of the active medium.

In Fig. 6, we present the results of the Caird analysis performed on the confocal resonator. The best fit resonator losses $\delta \approx 10\%$ are only slightly higher with respect to the previous cavity configuration, likely due to the insertion of the lens L_1 . As expected, the new resonator design was instrumental in significantly increasing the intrinsic slope efficiency to $\eta_0 \approx 49\%$.

To investigate the spatial beam quality, we collimated the laser output beam with a spherical lens $f_1 = 400$ mm, we focused the beam with a spherical lens $f_3 = 100$ mm, and we scanned the focal

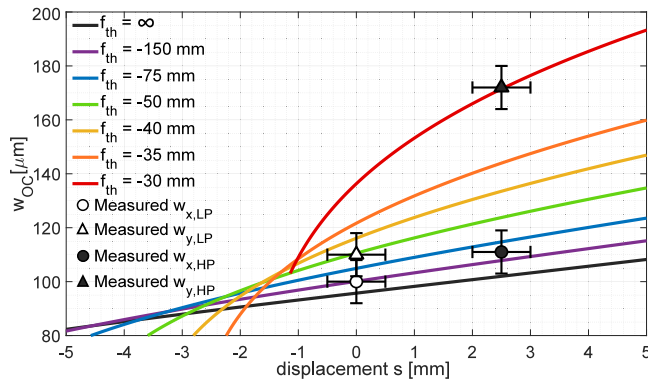


Fig. 7. Evolution of the confocal resonator fundamental mode beam waist radius on the OC as a function of the displacement s of mirror M_1 from the nominal distance of 178 mm from the pumped facet of the SCF and for different values of the focal length of the thermal lens in the SCF. Empty circle (triangle) corresponds to the measured w_{oc} along the x (y) direction of the lab reference at low P_{abs} (LP). Solid circle (triangle) corresponds to the measured w_{oc} along the x (y) direction of the lab reference at high P_{abs} (HP).

region with a CCD camera. In the inset of Fig. 6, we show the result of the measurement at the maximum absorbed pump power employing the optimum OC. As expected, the higher efficiency was obtained at an expense of a slightly worsened spatial beam quality with $M_x^2 \times M_y^2 \approx 2.3 \times 1.7$, and a slightly elliptical beam profile ($w_x \times w_y \approx 42 \mu\text{m} \times 57 \mu\text{m}$). The beam quality measurement was repeated for an absorbed pump power of 30 W, yielding a $M_x^2 \times M_y^2 \approx 1.3 \times 1.7$ and a beam waist in the focus $w_x \times w_y \approx 28 \mu\text{m} \times 36 \mu\text{m}$. From these data, through ABCD modeling of the cavity, we could estimate the focal length of the thermal lens f_{th} in the SCF. In Fig. 7 we show the evolution of the beam waist radius on the OC for different values of f_{th} as a function of the displacement s with respect to the nominal distance between the curved mirror M_1 and the pumped facet of the crystal (see Fig. 1). During the experiments, we observed that the optimization of the output power required a progressive translation of the curved mirror M_1 . In Fig. 7 we labeled with displacement $s = 0$ the initial condition, close to the threshold, corresponding to a distance between M_1 and the pumped facet of the SCF of ~ 178 mm. Back-tracing the embedded TEM_{00} mode waist on the OC through ABCD modeling of the beam quality measurement optical setup, we could estimate a strongly asymmetric thermal lens with focal length $f_{th,x} \approx -100$ mm and $f_{th,y} \approx -30$ mm, at the maximum absorbed pump power, as it is shown in Fig. 7.

According to the detailed investigation of thermal lensing effect in YLF crystal reported in [12], such stronger thermal lensing effect observed in the same pumping conditions as the previously described experiments can be a consequence of the reduction of the fundamental cavity mode dimension in the active medium in the confocal resonator, since the dioptric power of the thermal lens is strongly dependent on the on-axis displacement from the center of the pumped region.

3. Conclusion

In conclusion, we have presented for the first time, to the best of our knowledge, a laser resonator based on a birefringent Yb-doped SCF grown by the μ -PD technique. Under an incident pump power exceeding 100 W, we obtained up to 10.5 W in a diffraction limited beam quality out of a simple plane concave resonator. By enhancing the overlap efficiency between the pump and the cavity mode in a confocal resonator setup, we increased the output power to 25 W at an expense of a reduction of the laser beam quality to $M^2 \sim 2$. The Caird analysis we performed on the oscillators performance and the output beam quality we measured confirmed the good quality of the grown SCF. Thermal lensing effect was also investigated. For relatively

large resonating mode in the SCF, the thermal lens was almost spherical and its dioptric power was comparable to the one estimated in similar conditions in our previous investigations [8]. When the transverse dimension of the resonated mode was resized to enhance the oscillator efficiency, the latent anisotropy of thermo-optical properties of LLF crystal matrix was more evident. However, the fairly astigmatic and more severe thermal lensing did not affect significantly the laser beam quality.

The major limiting factor to the performance of the laser was likely the relatively large diameter (~ 2.8 mm) of the available SCF. However, despite this limitation and our SCF not being AR coated at the pump and laser wavelength, the results here reported very well compare in terms of output power, efficiency and beam quality with those obtained with Yb:YAG μ -PD grown SCFs at a similar stage of development of the technology [3]. We expect that by growing significantly thinner crystals we could be able to both circumvent the thermal degradation of the laser performance thanks to a reduction of the absolute temperature growth in the center of the SCF [2] and to exploit pump guiding effect in view of a potential pump power up-scaling of the very promising birefringent Yb-doped fluoride SCF.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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